AUTO HOME
PROJECT REPORT

B.Tech – Computer Science and Engineering

(CSE Core – SCOPE) Vellore Institute of Technology (VIT)

BACSE324 | Embedded Systems Architecture

Academic Year 2026–2027

Submitted by

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1. Team Members

S. No.	Name	Registration Number
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2. Objectives

- **RFID Authentication:** Use the RC522 reader to scan RFID tags and identify authorized users through their UIDs.
- **Access Control:** Grant or deny access based on predefined authorized UIDs.
- **Status Indication:** Use LEDs to indicate denied and granted access, respectively.
- **Motor Control:** Use an L293D motor driver and DC motor to demonstrate fan control.
- **Automatic Lighting**: Use an LDR to control the light indicator based on ambient brightness.
- **Temperature Monitoring:** Use an LM35 sensor to control AC indicators according to temperature thresholds.
- **System Testing:** Integrate and test all components for reliable operation under different conditions.

3. Working Principle

The project operates on the principles of electromagnetic coupling, analog sensing, digital processing, and electromechanical energy conversion. The Arduino Uno acts as the central processing unit, coordinating RFID authentication, environmental monitoring, and output control.
The system consists of three main stages:

Input stage: The RC522 RFID reader, RFID tags, LDR, and LM35 provide identification and environmental information.

Processing stage: The Arduino Uno processes the received UID, evaluates sensor readings against configured thresholds, and determines the required output states.

Output stage: LEDs indicate the authentication status, while the L293D motor driver controls the DC motor used to demonstrate a locking mechanism. Other outputs represent the operation of room lighting, a fan, and an AC.

The RFID subsystem handles access authorization, while the environmental sensors provide inputs for automatic room control. These functions can operate together, provided the software implements their interaction correctly.

3.1 Physics Involved

- **RFID:** The RC522 generates a 13.56 MHz electromagnetic field that powers the passive RFID tag. The tag transmits its UID through load modulation, which the reader decodes and sends to the Arduino via SPI.
- **LM35 Temperature Sensor:** Converts temperature into an analog voltage at approximately 10 mV/°C. The Arduino converts this voltage into a digital value using its ADC.
- **LDR:** Its resistance decreases as light intensity increases. A voltage divider converts this resistance change into a voltage that the Arduino measures.
- **L293D Motor Driver:** Uses an H-bridge circuit to control the motor's direction and operation based on

Arduino signals.

- **LEDs and Actuators:** LEDs indicate system status, while the motor demonstrates the locking and unlocking mechanism.

3.2 Algorithm / Program Logic

1. Initialize the RFID reader, sensors, LEDs, motor, and switches.
2. Scan an RFID tag and compare its UID with the authorized list.
3. If unauthorized, deny access and turn on the red LED.
4. If authorized, grant access, turn on the green LED, and activate the unlocking motor.
5. Read the LDR and LM35 sensor values to monitor light and temperature.
6. Control the room light, fan, and AC indicators according to predefined thresholds.
7. Repeat sensor monitoring while access is active.
8. Lock the system and return to RFID scanning when the session ends.

3.3 Data Flow Graph (DFG)

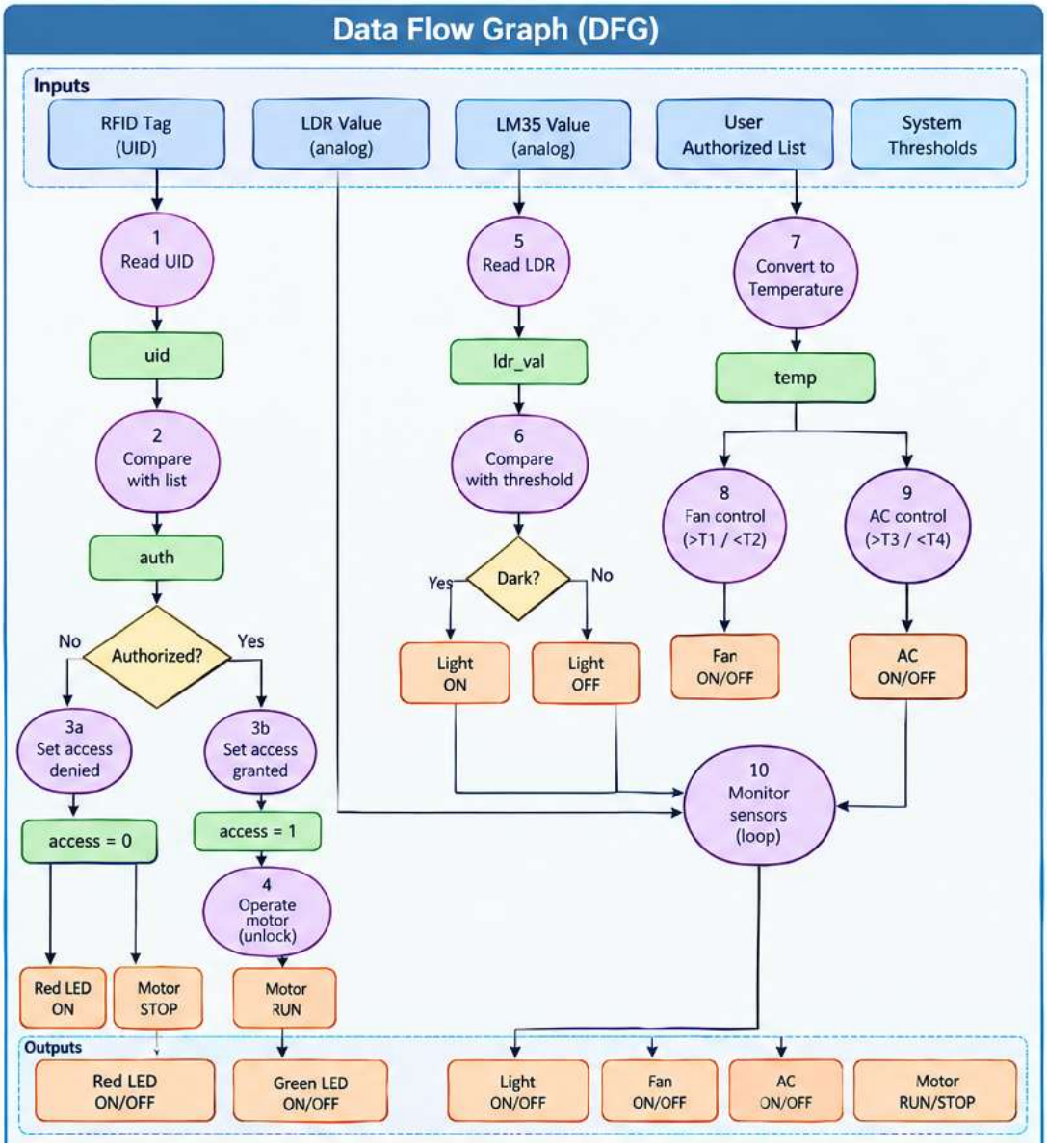


Figure 1. Data Flow Graph of the proposed system.

3.4 State Diagram (Moore)

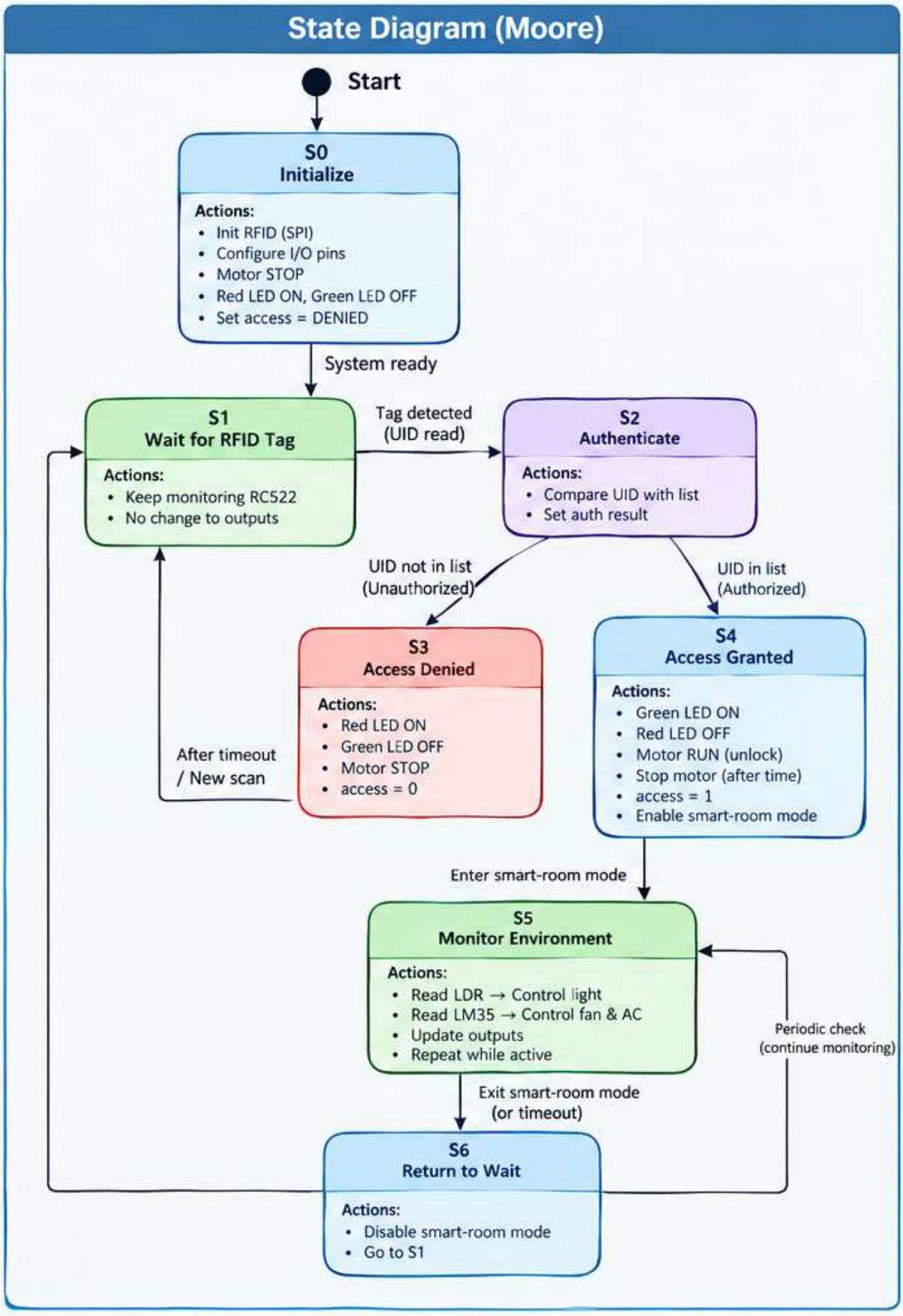


Figure 2. State diagram of the system.

4. Final Circuit Diagram

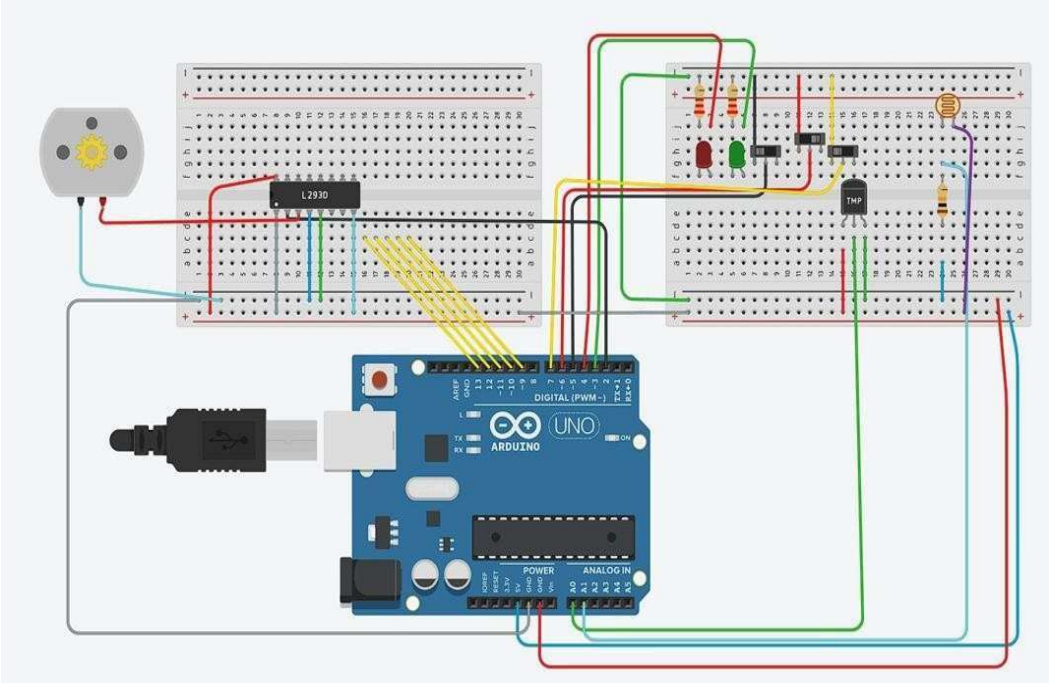


Figure 3. Final circuit diagram.

Arduino Uno connects to theRC522reader using the interface configured for the particular module and library. Red and green LEDs connect through current-limiting resistors. The DC motor connects through an L293D motor driver. All modules share a common ground.

(**NOTE** - TINKERCAD does not have RFID module so **5 yellow wires** are just connected to show connection with RFID reader)

5. Photograph of the Final Product

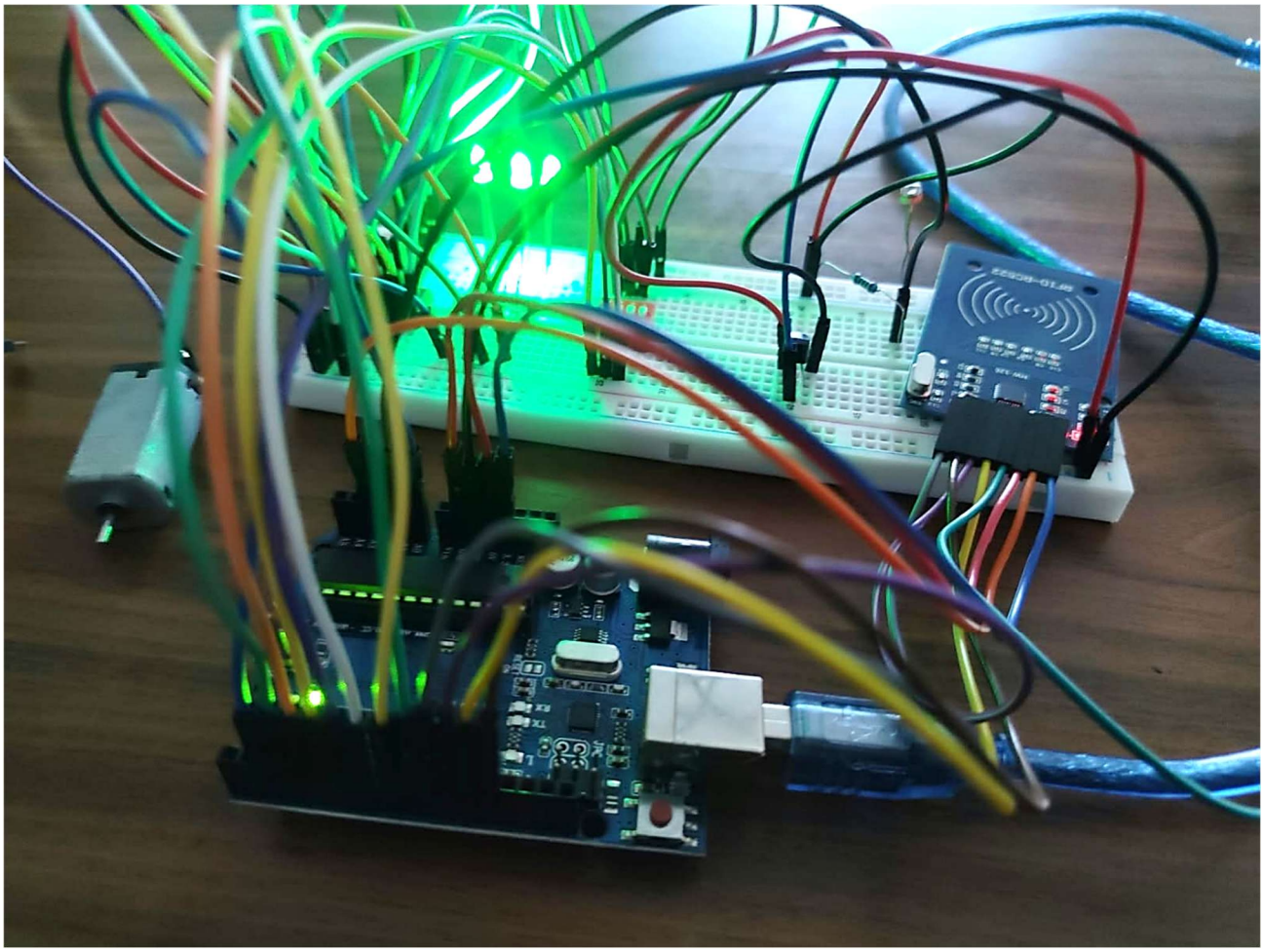


Figure 4. Photograph of the completed project.

6. Conclusion

1. Objectives Achieved:
Implemented RFID-based user authentication and designed sensor-based automation using an LDR for lighting control and an LM35 for temperature monitoring. Integrated the Arduino with the components required to control the light, fan, and AC.

2. Challenges Faced:
Ensuring proper communication between the RFID reader, sensors, and Arduino while maintaining reliable hardware connections. Configuring the sensor thresholds and output-control logic for accurate operation.

3. Improvements:
Improve wiring organization, calibrate the sensors, and refine the control logic for more reliable performance. Add suitable driver circuits and safety measures for controlling electrical appliances.

4. Future Scope:
Extend the system with mobile-app control, real-time room monitoring, and usage logging. Integrate IoT connectivity and occupancy detection to improve convenience and energy efficiency.

7. References

[1] NXP Semiconductors, “PN532 User Manual,” NXP Semiconductors. Use the revision corresponding to the module.

[2] Arduino, “Arduino Uno Rev3,” Arduino Documentation. https://docs.arduino.cc/hardware/uno-rev3/

[3] Arduino, “SPI Library,” Arduino Documentation.
https://docs.arduino.cc/language-reference/en/functions/communication/spi/

[4] Texas Instruments, “LM35 Precision Centigrade Temperature Sensors,” Texas Instruments. https://www.ti.com/product/LM35

[5] STMicroelectronics, “L293D Push-Pull Four-Channel Driver With Diodes,” STMicroelectronics. https://www.st.com/en/motor-drivers/l293d.html